

III. Other sense organs

- Olfactory sensillae** (a type of chemo receptors), which are sensitive to smell, are present on the **maxillary palps**, **labial palps** and **antennae**.
- Gustatory sensillae** (a type of chemoreceptors), which are sensitive to taste, are present on the **labrum**, **maxillary palps** and **labial palps**.
- Thermoreceptor sensillae** (receptors of temperature) occur on the 1st, 2nd, and 3rd segments of the 'tarsi' of the legs.

How is the overlapping of images prevented in diurnal insects?

7.9 REPRODUCTIVE SYSTEM OF PERIPLANETA

Periplaneta is dioecious, or unisexual, and both the sexes have well developed reproductive organs. The sexual dimorphism is evident both externally and internally. The female is different from the male in respect of short and broad abdomen, presence of brood pouch and absence of anal styles.

7.1.2 Male Reproductive System

The male reproductive system consists of a pair of **testes**. These are elongated and lobed structures lying one on each lateral side in the **fourth to sixth abdominal segments**. They are embedded in the fat bodies. From the posterior end of each testis, there starts a thin duct, the **vas deferens**. The two vasa deferentia run backwards and inwards to open into a wide median duct, the **ductus ejaculatorius** in the seventh segment. A characteristic **mushroom shaped gland** is present in the 6th and 7th abdominal segments which functions as an accessory reproductive gland.

The gland consists of two types of tubules: **i)** long slender tubules, the **utriculi majores** or 'peripheral tubules'. **ii)** Short tubules, the **utriculi breviores**. Secretion of utriculi majores forms the inner layer of the spermatophore while that of utriculi breviores nourishes the sperms. These tubules open into the anterior part of the **ejaculatory duct**.

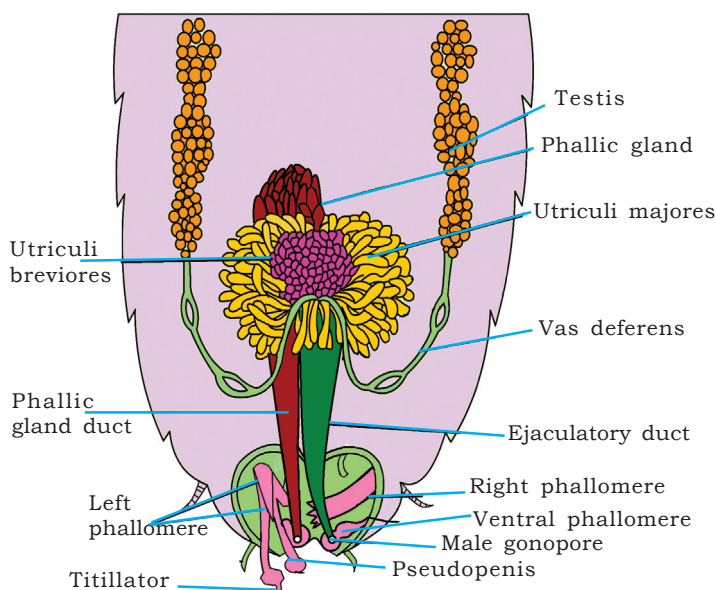


Fig. 7.9 *Periplaneta* : Male reproductive system

The **seminal vesicles** are present on the ventral surface of the ejaculatory duct. These sacs store the sperms in the form of bundles called **spermatophores**. The ejaculatory duct is a muscular tube that extends posteriorly and opens at the **gonopore** or the 'male genital pore' on the ventral phallomere. The duct of **phallic** or **conglobate gland** also opens near the gonopore. Its function is still not known. Surrounding the male genital opening there are three chitinous and asymmetrical structures called **phallic** organs or **gonapophyses** or **phallomeres** which help in copulation. These are the male external genitalia. The term phallomere refers to 'penis'. The left phallomere bears a pseudopenis and titillator.

Note: Ejaculatory duct is believed to secrete the middle layer and the conglobate gland secretes the outer layer of a spermatophore.

7.9.2 Female Reproductive System

The female reproductive system of *Periplaneta* consists of a pair of ovaries, a pair of oviducts, vagina, spermathecae, spermathecal papilla, and colleterial glands.

Ovaries

A pair of large **ovaries** lies laterally in 2 to 6 abdominal segments. They are light yellow in colour surrounded by fat bodies. Each ovary consists of eight tubules called ovarian tubules or **ovarioles**. Each ovariole consists of a tapering anterior filament called **germarium**, and a posterior wider **vitellarium**. The germarium contains various stages of developing ova, and the vitellarium contains mature ova with yolk. The tapering ends of the ovarioles of each ovary unite to form a single thread which attaches to the dorsal body wall. The ovarioles, at their posterior end unite to form a short wide **oviduct**. The oviducts unite to form a very short median **vagina**. The vertical opening of the vagina is called female genital pore. It opens into a large genital pouch on the eighth sternum. A spermatheca or '**receptaculum seminis**', consisting

of a left-sac like and a right filamentous caecum, is present in the 6th segment which opens by a median aperture on a small spermathecal papilla in the dorsal wall of the genital pouch on the ninth sternum. In a fertile female, the spermatheca contains spermatophores, received during copulation.

A pair of branched **colleterial glands** is present behind the ovaries. These glands open into the genital pouch separately, just above the

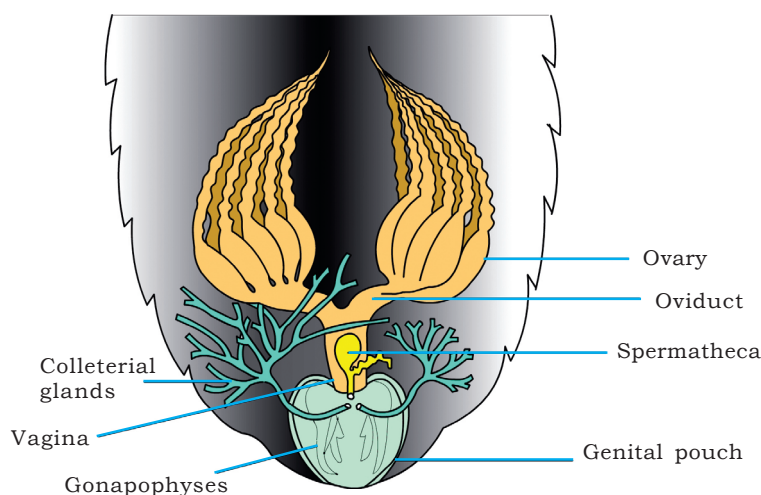


Fig. 7.9 *Periplaneta* : Female reproductive system

spermathecal aperture. Secretion of the two collateral glands forms a hard egg case called **ootheca** around the eggs.

Genital or brood pouch is formed by 7th, 8th, and 9th abdominal sterna. The sternum of the seventh segment is boat shaped and forms the floor and side walls of the genital pouch. The sterna of the eighth and ninth segments, which are tucked into the seventh segment, constitute the anterior wall and the roof of the genital pouch, respectively. The genital pouch has two chambers the anterior 'gynatrium' or **genital chamber** and posterior 'vestibulum' or **oothecal chamber**.

Three pairs of plate like chitinous structures called gonapophyses are present around the female genital aperture. These gonapophyses guide the ova into oothecal chamber acting as ovipositors. These are the *female external genitalia*, which help in copulation also.

7.9.3 Life history

The fertile female produces certain **pheromones** (chemical substances useful in chemocommunication), that attract the males for copulation. Copulation occurs during summer and it takes place during the night time. For copulation the male and female cockroaches bring their posterior ends very close. The male deposits a spermatophore on the spermathecal papilla of the female with the help of its phallomeres.

In *Periplaneta* sixteen eggs mature at a time and are passed into genital chamber. Fertilisation occurs in the gynatrium. The secretions of collateral glands form the egg case or **ootheca** around the fertilized eggs in the oothecal chamber. The ootheca is dark reddish to blackish brown capsule. It contains 16 eggs arranged in two rows of eight each. The oothecae are deposited at a suitable warm and dark place. On an average, female produces 9-10 oothecae. The development of eggs occurs inside the ootheca. Cockroach is **paurometabolous**, which means the development (metamorphosis) is gradual through nymphal stages. The eggs in ootheca undergo cleavage resulting in blastula followed by gastrula. Further development results in the formation of a **nymph**. It (nymph) is a young cockroach hatching from the ootheca. It resembles the adult, except in size, colouration, sexual immaturity and absence of wings. After hatching, the nymphs feed and undergo a series of 13 moultings or ecdyses. Shedding of exoskeleton is called '**moulting**' or '**ecdysis**'. After the last ecdysis the nymph grows into fully formed adult, with wings (the nymphs bear small wing pads).

Note: Moulting permits growth in the size of the nymph.

What do you name the young one of a cockroach which resembles the adult except in size, wings, colour and genitalia?

GLOSSARY

Abdomen: Third or posterior division (tagma) of an insect

Antenna: A sensitive feeler from the animal's head; tactile and olfactory in function

Appendage: A movable projecting part of the body; mouth parts are modified appendages.

Arolium: A soft hairy pad between claws in the leg of cockroach.

Arthropod: A phylum of segmented invertebrates with jointed legs.

Biramous: Possessing two branches e.g. the exopodite and endopodite.

Blastula: Embryonic stage with blastocoel, which is also called the 'primary body cavity'.

Cephalic: Relating or to the head.

Cervix: Neck.

Chitin: A polysaccharide-protein complex; a horny substance.

Copulation: Sexual process/union for the transfer of sperm from one to another of mating partner.

Cursorial: Swift runner.

Cuticle: External skeletal structure of the insect's body; it has epicuticle, mesocuticle and endocuticle.

Dioecious: Species population in which male and female organs occur in separate individuals.

Dioptric region: The part that focuses light rays on to the inner part of an ommatidium .

Ecdysis: Shedding of the outer body layer (cuticle).

Germarium: Tapering anterior filament of an ovariole.

Haemocoel: The body cavity of an arthropod or a mollusk filled with haemolymph; derived from the blastocoels of the embryo ; also called the 'primary body cavity'

Nervures: Tubular hollow network of the wing of a cockroach.

Ootheca: A hard purse like structure containing eggs in two rows.

Ovariole: A single tubule of an ovary containing developing eggs.

Paurometabolic: Gradual development/ metamorphosis by the nymphal stages of certain insects.

Podomere: A segment of an arthropod leg.

Pulvillus: A soft hairy pad between claws.

Sternum (plural: sterna): A ventral sclerite of a body segment of insects such as a cockroach.

Tagma (plural: tagmata): One of the regions of the body of an insect; division of the body into different regions is called tagmatisation.

Tergum (plural: terga) : Dorsal sclerite of the body segment of an insect.

VERY SHORT ANSWER TYPE QUESTIONS

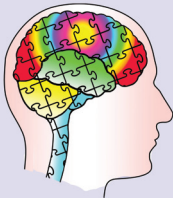
1. What are the structures with which cockroach walks on smooth surfaces and on rough surfaces respectively?
2. How is a tripod formed with reference to locomotion in cockroach?
3. Name the muscles that help in elevating and depressing the wings of a cockroach?
4. Name the different blood sinuses in cockroach?
5. How are the fat bodies similar to the liver of the vertebrates?
6. Which part of the gut secretes the peritrophic membrane in cockroach?
7. In which part of the gut of cockroach, water is reabsorbed?
8. Write the names of mouthparts in cockroach that help in biting and tasting the food.
9. What are alary muscles?
10. What is haemocoel?
11. Why is the blood of *Periplaneta* called haemolymph?
12. What is the function of haemocytes found in the blood of *Periplaneta*?
13. Write important functions of blood in *Periplaneta*?
14. What are trichomes? Write their functions.
15. Why is the respiratory system of cockroach called polypneustic and holopneustic system?
16. What is intima?
17. During inspiration which spiracles are kept open and which are kept closed?
18. Which factors regulate the opening of the spiracles?
19. The nitrogenous wastes in *Periplaneta* are removed from the body through alimentary canal. Why?
20. How do fat bodies help in excretion?
21. Which structure of cockroach acts as sensory and endocrine centre?
22. Distinguish between scolopidia and sensillae.
23. Which of the abdominal ganglia is the largest and why?
24. Why is the brain called the principal sensory centre in cockroach?
25. Distinguish between apposition image and superposition image.
26. List out the characters that help in understanding the difference between male and female cockroaches.
27. What is the function of mushroom gland in cockroach?
28. Compare the utriculi majores and utriculi breviores of the mushroom gland functionally.
29. What are gonapophyses?
30. How is colleterial gland helpful in reproduction of *Periplaneta*?

**SHORT ANSWER TYPE
QUESTIONS**

1. Draw a neat labeled diagram of the mouthparts of cockroach?
2. Describe the physiology of digestion in cockroach?
3. Draw a neat labeled diagram of the salivary apparatus of cockroach?
4. Describe the structure and function of the heart in *Periplaneta*?
5. Describe the process of blood circulation in *Periplaneta*?
6. What are different excretory organs in *Periplaneta*? Describe the process of excretion in detail.
7. How does *Periplaneta* conserve water? Explain it with the help of excretion in it?
8. Draw a neat and labeled diagram of ommatidium.
9. Describe the male reproductive system of cockroach.
10. Describe the female reproductive system of cockroach?

LONG ANSWER TYPE QUESTIONS

1. Describe the digestive system of cockroach with the help of a neat labelled diagram?
2. Describe the blood circulatory system of *Periplaneta* in detail and draw a neat and labelled diagram of it?
3. Describe the respiratory system of cockroach with the help of neat and labelled diagrams?
4. Describe the reproductive system of *Periplaneta* and draw neat and labeled diagrams of it.



For ignited minds

- NOT FOR EVALUATION

1. Do you know that animals send out nitrogenous wastes in the form of ammonia, urea, uric acid, guanine, etc. Ammonia is the **most toxic substance** and it can be sent only in very dilute form. It requires loss of more water (urine is hypotonic). Some fishes excrete ammonia (ammonotelic), as availability of water is not a problem for them. Urea is relatively less toxic and it can be sent out with relatively less water. Frog, earthworm etc., send out wastes mostly in the form of urea (ureotelic).

Note: Do you know that urea is formed mostly in our liver via the ornithine cycle? Uric acid is the least toxic substance and so it can be sent out in a concentrated form. So, animals such as reptiles, birds and insects for which conservation of water is very important send the wastes in the form of semisolid uric acid crystals (uric acid is insoluble in water, unlike ammonia and urea). Aquatic reptiles such as turtles excrete 'urea' like the mammals. However urine cannot be concentrated due to lack of Henle's loops in their nephrons. Now list out various types of animals depending on the nitrogenous waste excreted and state reasons for their nature.

2. When prawn, scorpion etc., have an oxygen carrying pigment in their haemolymph, why is it not seen in the blood of cockroach? Think of it.
3. When the labrum is removed or labral nerves are cut off, cockroach does not lose its ability/sense of tasting food. Why?
CLUE – Recall: functions of all appendages.
4. Why do insects moult in the early part of their life?
CLUE: It has something to do with growth.
5. The compound eye of cockroach, in a way, is a better adaptation for survival, than the human eye. Do you agree with the statement? If so how do you substantiate? **CLUE:** As a human being,

you can concentrate on only a particular part of your visual field only at a time, even though your visual field is much more larger/ wider.

6. What happens if the serially fifth abdominal ganglion is hypothetically damaged in a cockroach? The functioning of organs of which abdominal segment is affected?
7. What do you call the units of the mechanoreceptors in organ such as anal cerci?
8. If 20% of the ommatidia are damaged in each compound eye, a diurnal insect can see only 80% of the visual field. Do you agree with the statement? If so explain.
9. Why is storage of uric acid in corpora adiposa called storage excretion?
10. Of the following organs – antennae, mandibles, maxillae, legs and wings, which do you think is taxonomically significant when we think of insect such as a cockroach?
11. If the labro-frontal nerve is severed (cut off), will it affect the circulation of blood? How?
12. If the seminal receptacle of a cockroach is removed, and if copulation occurs soon after its removal what will be the effect on the reproductive ability of that cockroach?
13. The heart of a chordate is said to be myogenic (which means contraction is initiated in itself). If the heart is removed from the body it will keep beating provided some nourishing liquids such as Ringers solution is pumped through it. Will the same thing happen with the heart of the cockroach? Explain.
14. You know striated muscles appeared for the first time in the arthropods during evolution. Muscles need endo-skeletal parts for attachment. Cockroach does not have an endo-skeleton comparable to that of a chordate. Do you know how this problem of attachment surfaces is solved in cockroach? (Clue: Cuticle of cockroach shows inward extensions such as apodemes).
15. If the mesothoracic wings such as apodemes of a cockroach are removed, do you think it can still fly? If so, give reasons.